

skill gap calculation

1. Input skill score for each skill from user
2. **Formula**

For each skill:

Formula

For each skill:

$$\text{Gap}_i = W_i \times \frac{R_i - U_i}{R_i}$$

where:

- W_i = importance weight
- R_i = required proficiency
- U_i = user's proficiency (capped so $U_i \leq R_i$)
- If $U_i > R_i$, gap = 0 (extra proficiency doesn't reduce other gaps).

Overall skill gap:

$$\text{Skill Gap (\%)} = \left(\sum_{i=1}^n \text{Gap}_i \right) \times 100$$


Since the weights sum to 1, the maximum possible gap is 100%.

Example

Target Role: Data Scientist

Skill	Weight	Required	Student
Python	0.30	5	4
SQL	0.20	4	3
ML	0.25	5	2
Git	0.10	3	3
Docker	0.15	4	1

Step 1: Individual Skill Gaps

Python

$$0.30 \times (5 - 4 / 5) = 0.060.30$$

SQL

$$0.20 \times (4 - 3/4) = 0.050.20$$

ML

$$0.25 \times (5 - 2/5) = 0.150.25$$

Git

$$0.10 \times (3 - 3/3) = 00.10$$

Docker

$$0.15 \times (4 - 1/4) = 0.11250.15$$

Step 2: Total

$$0.06 + 0.05 + 0.15 + 0 + 0.1125 \\ = 0.3725$$

Step 3: Convert to Percentage

$$0.3725 \times 100 = 37.25\%$$

Final Skill Gap = 37.25%

Step 4 : calculate the gap for each category with different approach:

category	Required	Student
aptitude	5	4
soft skill	4	3
projects	5	2
internship	3	3

Aptitude:

$$(5-4) / 5 = 0.2$$

Soft Skill

$$(4-3) / 4 = 0.2$$

project :

$$(5-2) / 5 = 0.6$$

internship :

$$(3-3) / 3 = 0$$

Step 5: calculate overall gap

$$(\text{technical } w1) + (\text{projects } w2) + (\text{internship } w3) + (\text{aptitude } w4) + (\text{soft_skills} * w5)$$

example :

```
category,weight
Technical Skills,0.50
Projects,0.20
Internship,0.10
Communication,0.10
Aptitude,0.10
```

technical : 0.37

aptitude: 0.2

soft skill : 0.2

project : 0.6

internship : 0

$$(0.37 \ 0.5) + (0.6 \ 0.2) + (0 \ 0.10) + (0.2 \ 0.2) + (0.2 * 0.10)$$

$$0.185 + 0.12 + 0 + 0.04 + 0.02 = 0.365$$

$$0.365 * 100 = 36.5 \%$$

Skill Gap: 36.5 %

If student has no skill:

1. Overall gap:

$$0.50 + 0.20 + 0.10 + 0.10 + 0.10 = 1$$

$$\text{final gap} = 1 * 100 = 100\%$$

If student has all skills.

$$0 + 0 + 0 + 0 + 0 + 0 = 0 \%$$

Average gap:

$(\text{sum of all category gaps}) / (\text{total category}) * 100 = \text{average gap}$

if gap == 1 :

complete fresher

if gap == 0:

master (all skills)

if gap > 0 and gap <= 0.25:

skilled (great skills)

if gap > 0.25 and gap <= 0.50:

learner (intermediate skills)

if gap > 0.50 and gap <= 0.75:

beginner (basic skills)

if gap > 0.75 and gap < 0:

fresher (less skill)